Learned representations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Prepare the purpose, prerequisites and materials

> Prepare Learned representations: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Trace an **input** through a hidden **representation** and a decision rule

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Debugging shapes and gradients

Without opening the old lesson, explain one key idea from Debugging shapes and gradients. Give an example and a mistake that the idea helps you avoid.

Look for: Check dimension contracts before computing a layer

Reference example, not the only acceptable wording: Teacher explanation: No. Each weight row must have the same length as x.

Confirm the intended feature schema, then supply a valid two-feature **input** or deliberately redesign and retrain the layer; do not silently discard a value.

Regularisation and early stopping

Without opening the old lesson, explain one key idea from Regularisation and early stopping. Give an example and a mistake that the idea helps you avoid.

Look for: Separate a weight penalty from **prediction** error

Reference example, not the only acceptable wording: Teacher explanation: MSE 0, penalty 2, total J 2, total gradient 2. Zero data gradient does not remove the penalty gradient.

Batching and epochs

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish an example a batch an update and an epoch

Reference example, not the only acceptable wording: Teacher explanation: Batch lengths 4, 4, 2; three updates per epoch, nine total updates and 30 example visits.

Use two 45-minute sessions. Begin with rectangle measurements and the explicit strict comparison. Distinguish the constructed layer from a genuinely fitted layer throughout.

The key reasoning is the **collision** argument: identical **representation**s force identical deterministic **output**s. No calculus, external AI account or full **training** framework is required.

Outcomes

Trace an **input** through a hidden **representation** and a decision rule

Identify distinctions that a **representation** preserves or loses

Separate a useful constructed feature from **evidence** of successful learning

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

Different negative scores all become zero. Matching activations can hide an earlier error, so intermediate values need independent checks.

End goal for the pictorial case

By the end, I can explain learned **representation**s using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain and model the mechanism

> Model Learned representations using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Explain Learned representations · 1

Consider fictional rectangles with **input** [width, height] in cm. The task is to identify rectangles whose width is strictly greater than height. The target is 1 for wider rectangles and 0 otherwise.

A square therefore has target 0. This is an arithmetic teaching task, not a claim that a network is needed to measure rectangles.

Define two hidden units: h1 = ReLU(width - height) and h2 = ReLU(height - width). Their weight rows are [1, -1] and [-1, 1], with zero biases. The **representation** h = [h1, h2] records excess width or excess height.

Predict 1 when h1 > 0; otherwise predict 0.

These weights were chosen by the lesson author. They demonstrate a **representation** that could be implemented by a network, but no **training** run has established that a model learned it.

Learning would require adjusting parameters using **training** data and recording the procedure.

A **prediction** head is the computation that reads the hidden vector and produces an **output**. Keeping it separate lets us inspect what information is available before making a decision.

This example assumes two finite, positive measurements in cm; it is not a general **input** validator.

The raw difference width - height can be recovered as h1 - h2. The original width and height cannot in general be recovered: many rectangles share a difference.

Two hidden components do not guarantee preservation of two **input** components.

Term	Meaning
Representation	An encoding of an example used by another computation
Hidden activation	A value produced inside a network after applying an activation function
Collision	Two different inputs receiving the same representation

Term	Meaning
Learned feature	A feature whose computation depends on parameters adjusted using training data

Different rectangles can share a hidden vector

INPUT IN CM	HIDDEN VECTOR	AREA IN SQUARE CM
Width 4 height 2	[2, 0]	8
Width 5 height 3	[2, 0]	15

The representation preserves the wider-than-height answer for these examples but discards information needed to recover their different areas. A deterministic head receiving only this vector must give both examples the same output.

Explain Learned representations · 2

Guided: [6, 4] and [7, 5] both encode as [2, 0], predict 1 and have areas 24 and 35 square cm. A deterministic head using only h cannot recover both exact areas.

Exit: hand-chosen weights are not **evidence** of learning, and correct comparison labels do not establish area **prediction**. Test the **collision** [4, 2] versus [5, 3], with areas 8 and 15.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Different rectangles can share a hidden vector

INPUT IN CM	HIDDEN VECTOR	AREA IN SQUARE CM
Width 4 height 2	[2, 0]	8
Width 5 height 3	[2, 0]	15

The representation preserves the wider-than-height answer for these examples but discards information needed to recover their different areas. A deterministic head receiving only this vector must give both examples the same output.

Encode [6, 4] and [7, 5]. Give their wider-than-height predictions and areas. Can a head that sees only h always recover the exact area? Explain using this pair.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Learned representations

See when a **representation** preserves a distinction and when it destroys it.

The facts we will use

Supplied facts

Toy **input** [left, right]; **representation** h=[ReLU(left-right), ReLU(right-left)]. A=[5, 2]; B=[2, 5]; target is left-brighter versus right-brighter.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/19

A representation can preserve direction

	left	right	h1	h2
A	5	2	3	0
B	2	5	0	3

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/19

Learned representations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Encode A

A becomes [3, 0], with left brighter.



2. Encode B

B becomes [0, 3], with right brighter.



3. Compare collapsed summary

If the head receives only h1+h2, both become3 and the target distinction is lost.

CHANGE ONE THING

The head keeps both ordered hidden values instead of their sum.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Encode A	A becomes [3, 0], with left brighter.
2. Encode B	B becomes [0, 3], with right brighter.
3. Compare collapsed summary	If the head receives only h1+h2, both become3 and the target distinction is lost.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A hidden **representation** decides which distinctions remain available to an **output** head. These two hidden coordinates preserve whether the left or right side is brighter.

Adding them discards that direction and gives both examples the same scalar. A deterministic head cannot assign different labels to identical **input**s; retain the distinction or change the task.

What this example does not establish

Keep the stated limit

This is a hand-chosen **representation**, not **evidence** that a trained image model actually

learned these units.

Change one thing in the pictured case

Change one thing

Change: The head keeps both ordered hidden values instead of their sum.

Predicted result

Prediction: It can distinguish [3, 0] from [0, 3] in this example.

Why the result follows

Reason: The **representation** retains the direction of brightness difference when coordinates stay separate.

Ask while showing the picture

Ask while showing the picture · 3

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a **representation**; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Learned representations, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve how ReLU can hide distinctions and establish the rectangle schema and target.
7 to 18 minutes	Trace the four lesson examples, including a square, before naming the hidden features.
18 to 32 minutes	Complete W1 and trace the optional code, separating the representation from its head.
32 to 45 minutes	Use the visual and W2 to explain why identical hidden inputs cannot yield two exact areas.
Session 2 start to 8 minutes	Complete the guided pair and compare the sum-only encoding using W3.
8 to 20 minutes	Derive recovery from sum and difference, then discuss the limited conclusion in W5.
20 to 35 minutes	Collect W4 independently, requiring both score vectors, hidden vectors and target labels.
35 to 45 minutes	Use the exit response and fresh retry to distinguish learning evidence from an authored formula.

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide paper rectangles and columns for signed difference, two scores, activations and target. Allow calculators and ask for verbal **collision** explanations before algebra.

Extension: show that adding the same positive amount to width and height preserves h while changing area; derive exact recovery when the sum is also retained.

Reduce help deliberately

Completion practice

Completion task: Give the hidden vectors and areas for [4, 2] and [5, 3]. Explain why increasing only the complexity of a deterministic area head cannot make both exact **prediction**s correct.

Reasoning to listen for

Teacher check: Teacher explanation: Both hidden vectors are [2, 0], while areas are 8 and 15 square cm. Identical **input**s to a deterministic head yield identical **output**s, so the head cannot return both required values.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Learned **representation**s. Proposed explanation: “Two hidden values preserve both **input** measurements”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Dimension alone does not guarantee recovery; distinct **input**s can collide.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

Both examples supply the identical scalar3 but require different targets. Identical **input**s cannot produce different deterministic **output**s. Changed-case check: It can distinguish [3, 0] from [0, 3] in this example.

The **representation** retains the direction of brightness difference when coordinates stay separate.

Changed-case reasoning: It can distinguish [3, 0] from [0, 3] in this example. The **representation** retains the direction of brightness difference when coordinates stay separate.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the investigation and diagnose misconceptions

> Run the Learned representations investigation and use the first unsupported step to choose a teaching response.

Two hidden values preserve both **input** measurements: Dimension alone does not guarantee recovery; distinct **input**s can collide.

A more complex head can always fix lost information: A deterministic head cannot distinguish identical **representation**s without additional **input**.

Correct **output**s prove human concepts were learned: The layer here was constructed, and a task score alone does not establish a hidden concept.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Learned representations questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Guided: [6, 4] and [7, 5] both encode as [2, 0], predict 1 and have areas 24 and 35 square cm. A deterministic head using only h cannot recover both exact areas.

Exit: hand-chosen weights are not **evidence** of learning, and correct comparison labels do not establish area **prediction**. Test the **collision** [4, 2] versus [5, 3], with areas 8 and 15.

W1 Trace new inputs

Workbook question · W1

Encode [8, 3], [3, 8] and [4, 4] with the lesson layer. Apply its **prediction** rule and explain the square boundary.

Teacher reasoning · W1

Teacher explanation: The vectors are [5, 0], [0, 5] and [0, 0]. **Prediction**s are 1, 0 and 0. Equal width and height does not satisfy strictly greater.

W2 Prove an information loss

Workbook question · W2

Give the hidden vectors and areas for [4, 2] and [5, 3]. Explain why increasing only the complexity of a deterministic area head cannot make both exact **prediction**s correct.

Teacher reasoning · W2

Teacher explanation: Both hidden vectors are [2, 0], while areas are 8 and 15 square cm. Identical **input**s to a deterministic head yield identical **output**s, so the head cannot return both required values.

W3 Compare another encoding

Workbook question · W3

The alternative **representation** is s = width + height. Use [4, 2] and [2, 4] to decide whether s alone always supports the wider-than-height task.

Teacher reasoning · W3

Teacher explanation: Both have s = 6, but their targets are 1 and 0. This **collision** prevents any deterministic head using only s from getting both right.

W4 Audit a new representation independently

Workbook question · W4

A different layer uses g = [ReLU(width - height - 1), ReLU(height - width - 1)]. Encode [4, 3] and [3, 4]. State their targets for width strictly greater than height.

Can any deterministic head receiving only g classify both correctly?

Teacher reasoning · W4

Teacher explanation: Both encode as [0, 0]: the scores are [0, -2] and [-2, 0]. Targets are 1 and 0. No deterministic head receiving only this shared vector can distinguish the pair.

The added threshold erased a distinction required by this task.

W5 Repair and qualify a claim

Workbook question · W5

For h = [2, 0] and s = 8, recover width, height and area. Does this algebra prove that a **training** program learned the **representation** or that it will work well on noisy measurements?

Teacher reasoning · W5

Teacher explanation: Difference d is 2. Width is (8 + 2) / 2 = 5 and height is (8 - 2) / 2 = 3; area is 15 square cm.

The algebra establishes recovery under the formula, not a **training** history or performance with noisy measurements.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4 requires scores [0, -2] and [-2, 0], hidden vector [0, 0] for both, targets 1 and 0 and an impossibility explanation. Fresh retry: use g = [ReLU(width - height - 2), ReLU(height - width - 2)] for [5, 3] and [3, 5].

Scores are [0, -4] and [-4, 0]; both hidden vectors are [0, 0], but targets are 1 and 0. No deterministic head using only g can classify both correctly.

Assess independent transfer

Independent question · withhold the key

Selected case: A different layer uses g = [ReLU(width - height - 1), ReLU(height - width - 1)]. Encode [4, 3] and [3, 4]. State their targets for width strictly greater than height.

Can any deterministic head receiving only g classify both correctly?

Selected case answer

Teacher key · Selected case answer

Teacher explanation: Both encode as [0, 0]: the scores are [0, -2] and [-2, 0]. Targets are 1 and 0. No deterministic head receiving only this shared vector can distinguish the pair.

The added threshold erased a distinction required by this task.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect the project, support and next lesson

> Connect Learned representations to Image-pattern Network Investigation, adapt support and plan a later changed-case check.

Add a labelled learned **representation**s **evidence** sheet to Image-pattern Network Investigation. Show how you can trace an **input** through a hidden **representation** and a decision rule.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide paper rectangles and columns for signed difference, two scores, activations and target. Allow calculators and ask for verbal **collision** explanations before algebra.

Extension: show that adding the same positive amount to width and height preserves h while changing area; derive exact recovery when the sum is also retained.

Week 20 introduces image tensors and channels, with explicit axis order and value scales.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/20, ai-5/22, ai-5/25. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Learned representations lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Trace an **input** through a hidden **representation** and a decision rule

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Give the hidden vectors and areas for [4, 2] and [5, 3]. Explain why increasing only the complexity of a deterministic area head cannot make both exact **prediction**s correct.

Supported practice answer

Reasoned practice answer: Teacher explanation: Both hidden vectors are [2, 0], while areas are 8 and 15 square cm.

Identical **input**s to a deterministic head yield identical **output**s, so the head cannot return both required values.

Check every learner before discussion: Use the worked case for Learned **representation**s. Proposed explanation: “Two hidden values preserve both **input** measurements”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Dimension alone does not guarantee recovery; distinct **input**s can collide.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: The head keeps both ordered hidden values instead of their sum. Require a **prediction** and a reason before comparing. Expected result: It can distinguish [3, 0] from [0, 3] in this example.

Reason: The **representation** retains the direction of brightness difference when coordinates stay separate.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Learned representations graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Learned representations: the case

AI ACADEMY / AI-5/19 · 01 · LOOK AT THE FACTS

Learned representations: the case

The situation

See when a representation preserves a distinction and when it destroys it.

Supplied evidence

Toy input [left, right]; representation h=[ReLU(left-right), ReLU(right-left)]. A=[5, 2]; B=[2, 5]; target is left-brighter versus right-brighter.
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain learned representations using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

See when a **representation** preserves a distinction and when it destroys it. Supplied facts: Toy **input** [left, right]; **representation** h=[ReLU(left-right), ReLU(right-left)].

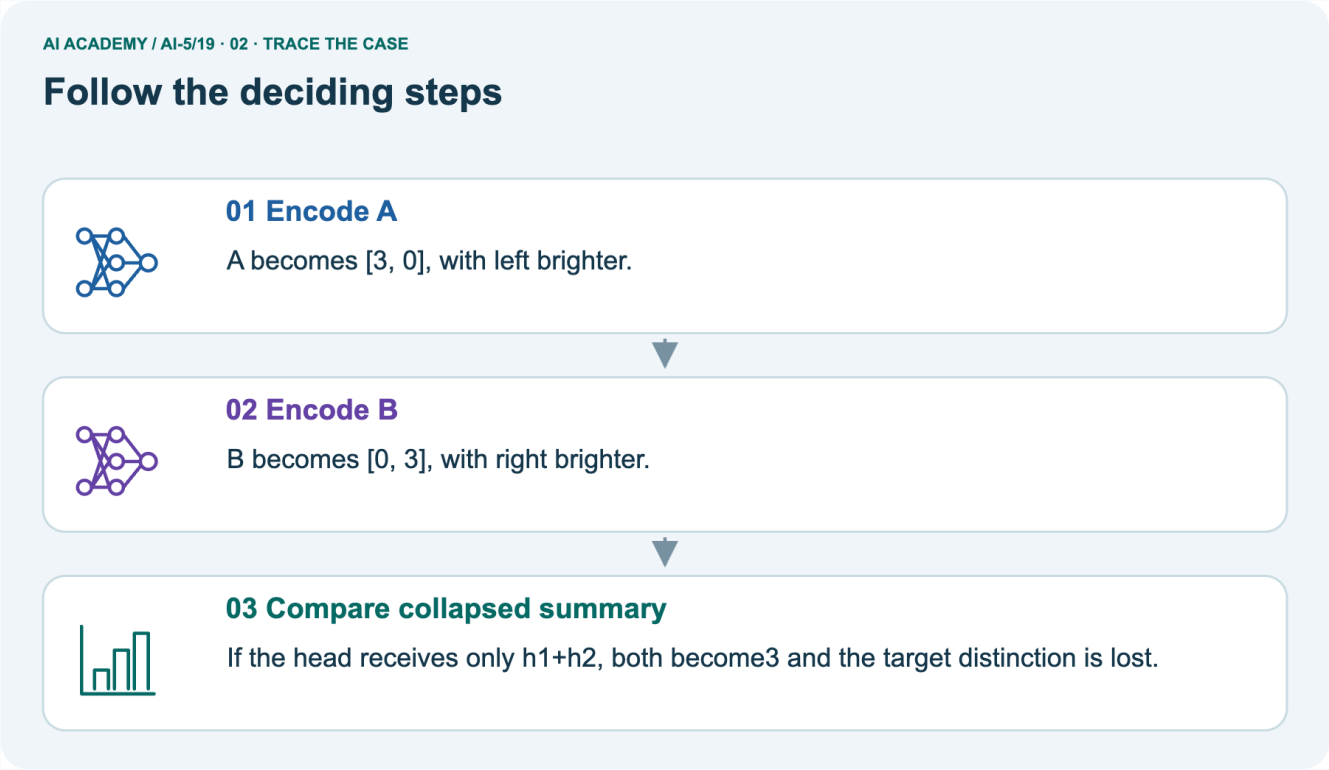
A=[5, 2]; B=[2, 5]; target is left-brighter versus right-brighter. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

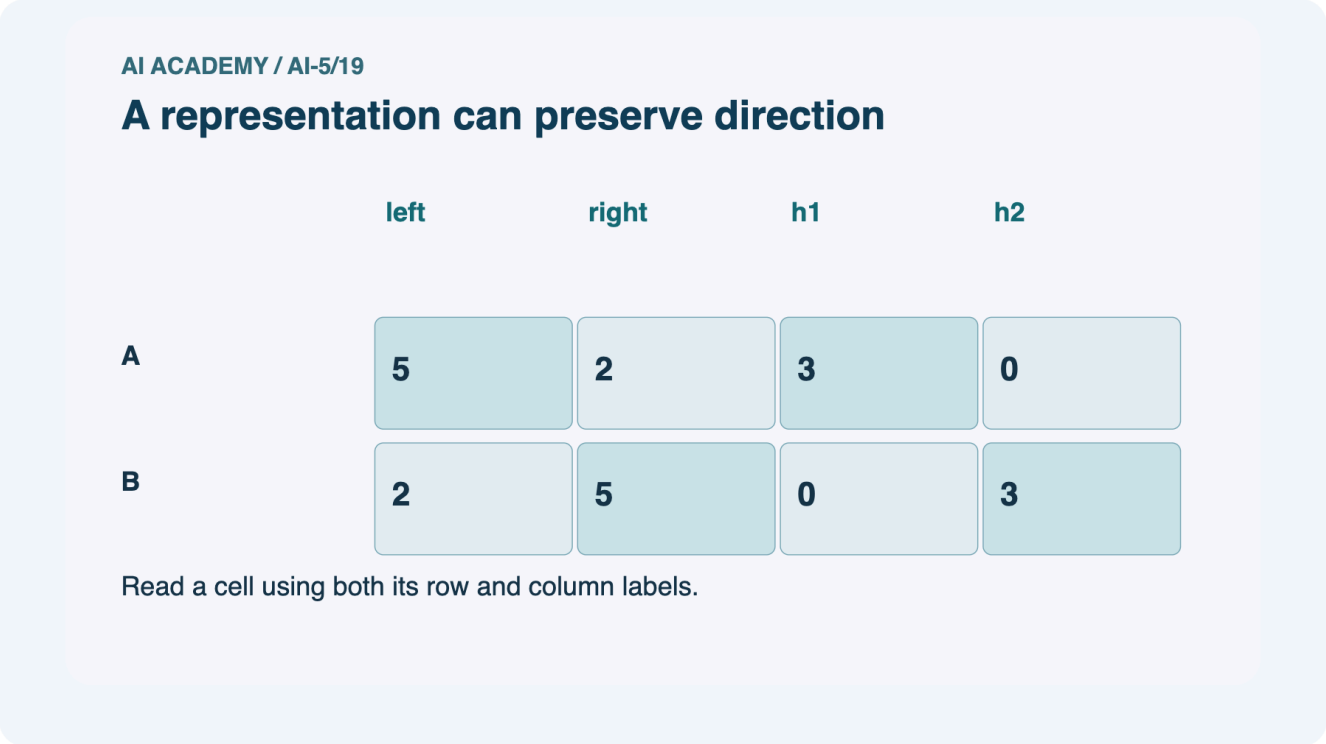
Encode A: A becomes [3, 0], with left brighter. → Encode B: B becomes [0, 3], with right brighter. → Compare collapsed summary: If the head receives only h1+h2, both become3 and the target distinction is lost.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. A representation can preserve direction



Follow the labelled picture with the matching explanation. Open the vector diagram

A **representation** can preserve direction. A: left = 5, right = 2, h1 = 3, h2 = 0. B: left = 2, right = 5, h1 = 0, h2 = 3.

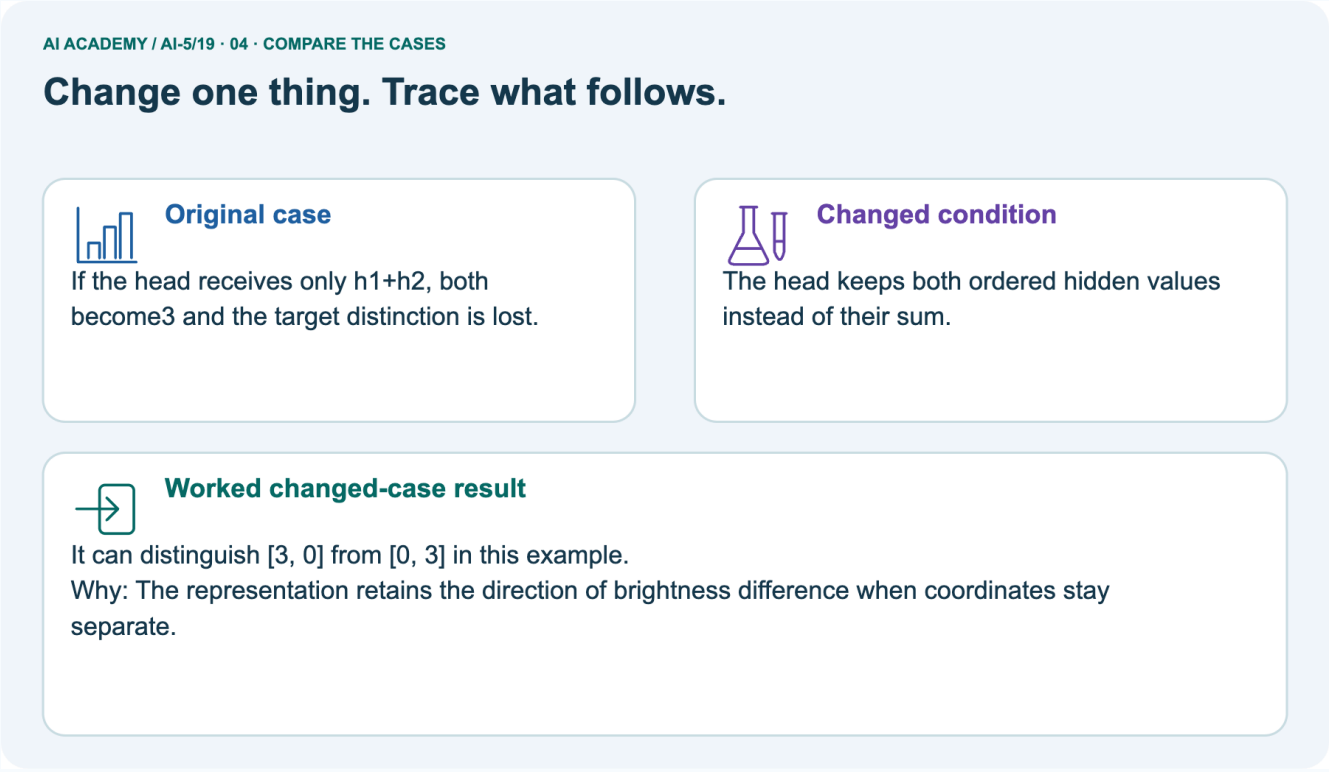
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: If the head receives only h1+h2, both become3 and the target distinction is lost. Changed condition: The head keeps both ordered hidden values instead of their sum.

Expected result: It can distinguish [3, 0] from [0, 3] in this example. Reason: The **representation** retains the direction of brightness difference when coordinates stay separate.

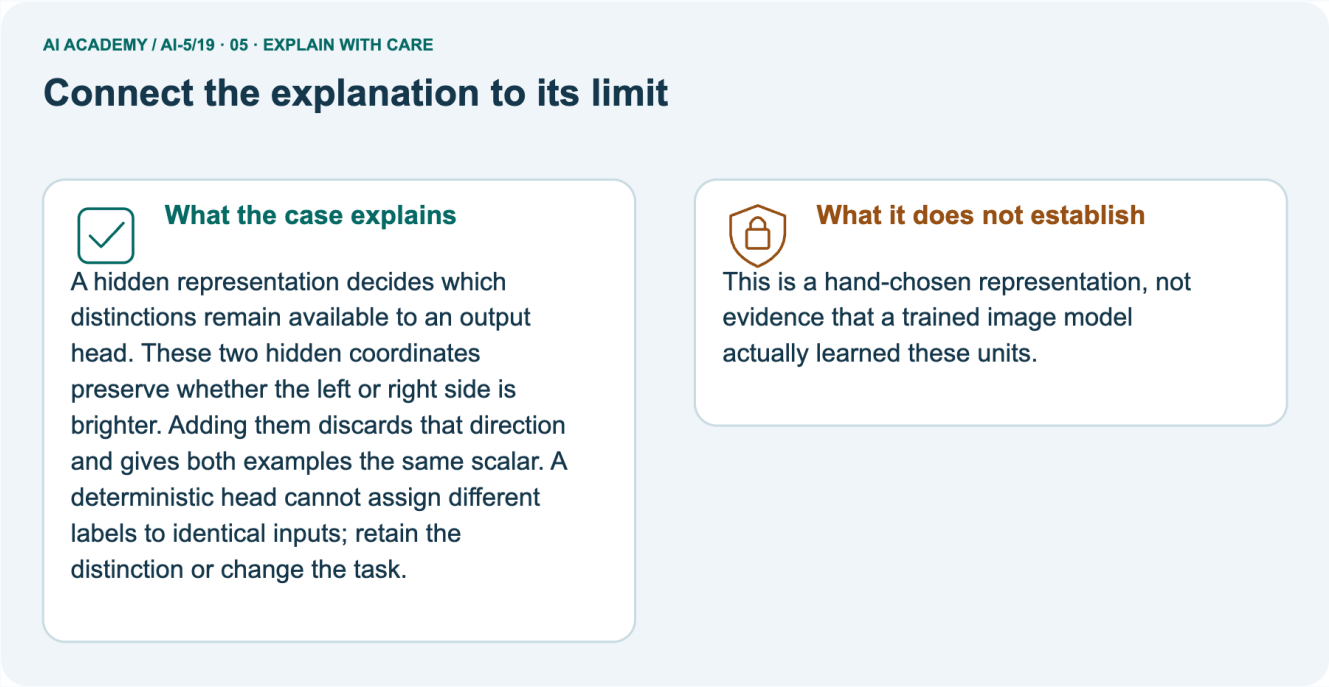
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The **representation** retains the direction of brightness difference when coordinates stay separate.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A hidden **representation** decides which distinctions remain available to an **output** head. These two hidden coordinates preserve whether the left or right side is brighter.

Adding them discards that direction and gives both examples the same scalar. A deterministic head cannot assign different labels to identical **input**s; retain the distinction or change the task.

Limit: This is a hand-chosen **representation**, not **evidence** that a trained image model actually learned these units.

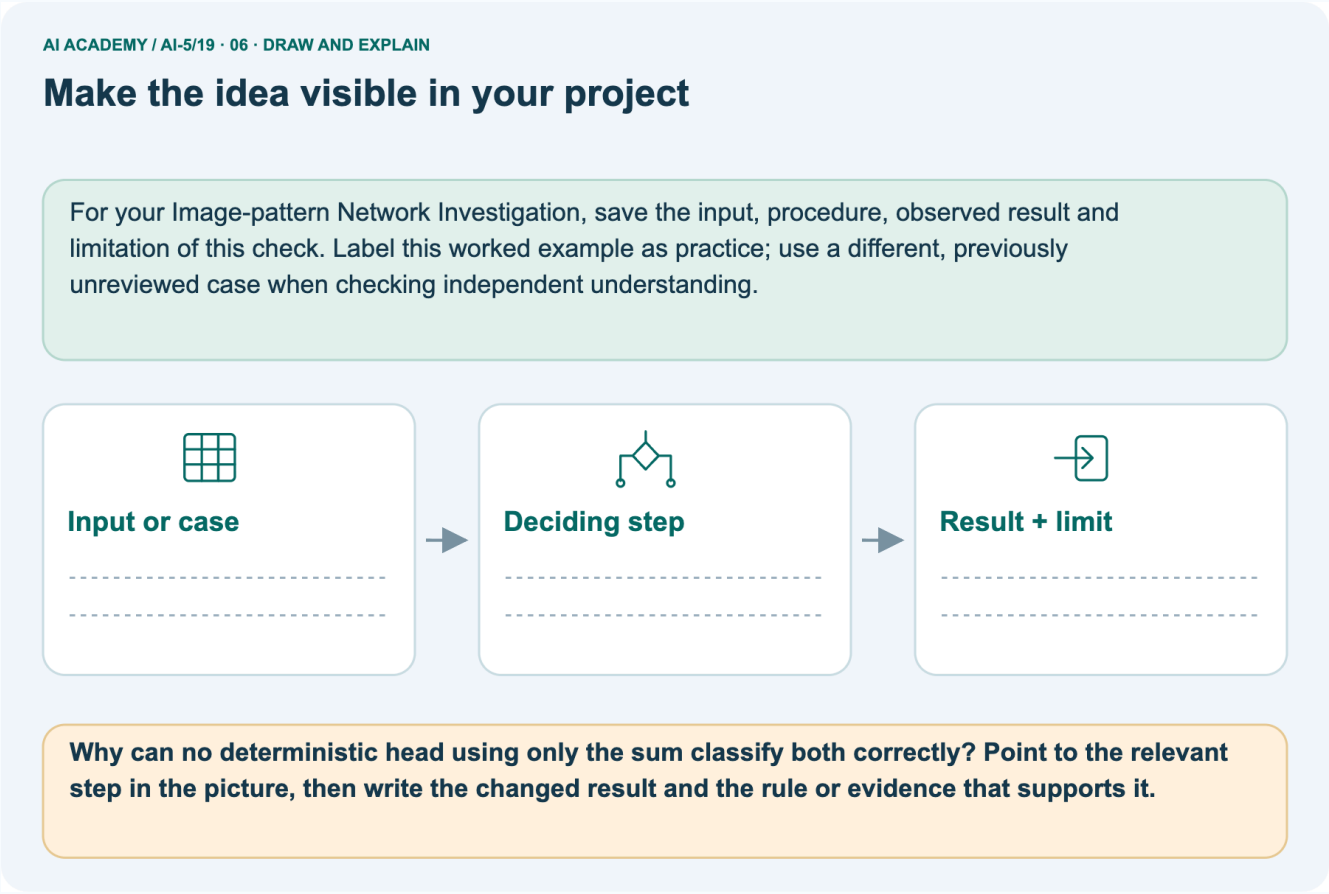
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This is a hand-chosen **representation**, not **evidence** that a trained image model actually learned these units.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Both examples supply the identical scalar3 but require different targets. Identical **input**s cannot produce different deterministic **output**s.

Changed-case check: It can distinguish [3, 0] from [0, 3] in this example. The **representation** retains the direction of brightness difference when coordinates stay separate.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.